Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EE7204 / EC7205

Project Title

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1. Problem Statement

Ground Glass Opacities (GGOs) are faint cloudy areas seen in lung CT scans. They can be early signs of diseases such as pneumonia, lung infections, or even cancer. Finding these areas by hand is difficult and takes a lot of time because GGOs have low contrast and spread out gradually in the lungs. Different doctors may also interpret them differently, which can affect diagnosis accuracy.

Traditional methods do not have clear and standard ways to measure GGOs, Because of this, it is very hard to accurately track how a disease changes over time or to compare scans taken before and after treatment.medical images stored in DICOM format are often handled inefficiently, windowing settings (brightness and contrast) vary between hospitals, and many steps are still done manually instead of automatically. These issues make it difficult to interpret CT scans in a consistent and standardized way.

2. Objectives

- Improve the quality of the chest CT scans by removing the noise, normalizing the image intensity values, and converting the data into a format suitable for processing.
- To accurately segment lung regions using HU-based thresholding, morphological operations, and edge filling techniques while removing background artifacts.
- To detect Ground Glass Opacity regions within the segmented lung area using HU based thresholding criteria.
- Test the efficiency of the proposed image processing workflow on a publicly available chest CT dataset.
- Evaluate the effectiveness of the proposed image processing pipeline using a publicly available chest CT dataset.
- Design and show a simple prototype application which focuses on clear image visualization and the ease of interpretation of the results.

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3. Literature Review

Study 1: Texture based adaptive multiple feature method for pulmonary parenchyma evaluation [1]

Introduced an adaptive multiple feature method (AMFM) using multiple statistical and fractal texture features extracted from CT images to quantify emphysema and related parenchymal abnormalities. The approach divided slices into regions (ventral to dorsal) and applied texture analysis to discriminate normal from abnormal tissue, outperforming simpler mean lung density or histogram percentile methods. It achieved high accuracy (up to 100% globally, ∼98% regionally) in separating normal and emphysematous areas. Strengths included objective, quantitative assessment without heavy reliance on fixed thresholds and good regional discrimination. Limitations were its focus on emphysema rather than GGO specifically, computational demands for texture computation at the time, and sensitivity to slice orientation or scanner differences. Relation to our project: We extend intensity and texture concepts to GGO specific ranges (-700 to -500 HU) with added morphological refinement and 3D reconstruction for better spatial context, while keeping the pipeline computationally efficient and interpretable.

Study 2: Region growing and morphological operations for abnormality isolation in CT [2]

Approaches in this era used seeded region growing within initial lung masks (often from thresholding) to isolate ground glass attenuation areas, combined with morphological opening and closing to smooth boundaries and remove noise. These methods correlated CT patterns like ground glass opacification with functional abnormalities in conditions such as hypersensitivity pneumonitis. Strengths lay in improved handling of fuzzy edges and noise compared to pure thresholding, plus integration of local context for better abnormality delineation. Limitations included parameter sensitivity (seed selection, growth criteria) leading to variability across scans, reduced performance on diffuse or faint GGO, and limited 3D volumetric processing. Relation to our project: Our pipeline incorporates connected component analysis post thresholding and advanced morphological operations (erosion, dilation, closing) for robust lung and GGO mask refinement, addressing initialization issues through adaptive HU windows and adding true 3D volume rendering for enhanced visualization.

Study 3: Hybrid lung segmentation with error detection and morphological refinement [3]

Hybrid methods combined rule based segmentation (e.g., thresholding and edge detection) with pixel classification or morphological post processing to segment lung fields, often in chest radiographs but adaptable to CT. These included error detection mechanisms to switch strategies when initial segmentation failed. Strengths were robustness to variations (noise, pathology) and high accuracy (~94–97%) through complementary techniques. Limitations involved complexity in hybrid rules, potential for residual errors in severe abnormalities, and less emphasis on GGO specific sub segmentation or interactive 3D views. Relation to our project: We adopt a similar hybrid philosophy starting with HU thresholding for lung extraction, followed by connected components and morphology but extend it to full volumetric 3D processing with GGO candidate highlighting and interactive overlays, prioritizing transparency over classification-heavy hybrids.

Study 4: Markov random field-based segmentation for GGO in lung CT [4]

Probabilistic frameworks using Markov random fields modeled pixel relationships after initial thresholding (e.g., −374 HU or adjusted) and morphological lung boundary refinement (median filtering, opening/closing, erosion). Energy minimization labeled GGO regions with high sensitivity/specificity (~87–94%). Strengths included effective handling of fuzzy boundaries and noise reduction via probabilistic context. Limitations were computational cost for optimization, need for parameter tuning, and challenges with very subtle or heterogeneous GGO patterns. Relation to our project: While we avoid probabilistic optimization for speed and simplicity, we incorporate similar preprocessing (noise filters, morphology) and HU-based candidate extraction, enhanced by local texture variance checks and interactive 3D pseudo-color rendering to improve clinical interpretability without added complexity.

Study 5: Threshold-based GGO candidate detection with density masks [5]

Density mask techniques applied fixed HU thresholds (e.g., −750 to −300 HU) to quantify opacities, sometimes compared to neural networks for sensitivity gains. These methods segmented GGO areas post-lung isolation and reported high sensitivity but lower specificity due to false positives. Strengths were extreme efficiency and direct radiological alignment (HU-based). Limitations included noise susceptibility, partial volume effects, and poor specificity without refinement, often detecting 17%+ extraneous areas. Relation to our project: We use targeted HU ranges (−700 to −500) for GGO but add morphological false-positive reduction and texture discrimination, plus full 3D interactive visualization with overlays addressing specificity and usability gaps in basic density approaches.

Study 6: Semi-automated GGO segmentation with radiomics and threshold tools [6]

Semi-automated pipelines used threshold tools (e.g., −1350 to −700 HU) in software like 3D Slicer, followed by manual correction and morphological/logical operations for lung/GGO ROIs. These supported radiomics feature extraction for differentiation (e.g., COVID-19 vs. other diseases). Strengths included high control and integration with quantification tools. Limitations were partial automation, user dependency, and less focus on fully interactive 3D prototypes. Relation to our project: Our fully automated yet editable pipeline (with annotation/measurement tools) builds on threshold foundations but emphasizes end-to-end interactivity, 3D rendering, and visual explainability without requiring manual ROI tweaks for initial results.

Gap Analysis and Project Contributions

Existing works excel in foundational thresholding, morphological refinement, and texture analysis for lung/GGO tasks but often lack seamless 3D interactive visualization, color-coded opacity highlighting, and user-friendly annotation/measurement in one tool. Many remain 2D-focused or parameter-sensitive without strong emphasis on reproducibility and clinician control. Our project integrates these classical CV/IP elements ($> 65\%$ focus) into a comprehensive prototype: DICOM preprocessing, adaptive lung segmentation, precise GGO detection with texture/morphology filters, and interactive 3D volume rendering with overlays. This advances interpretability, accessibility, and practical utility for monitoring subtle abnormalities on public datasets.

4. Methodology

(Minimum word limit: 1000 words) Describe the approach, tools, and algorithms you will use. Mention the main image processing or computer vision techniques (e.g., segmentation, feature extraction, classification) and the software frameworks (e.g., OpenCV, MATLAB, TensorFlow).

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5. Dataset / Data Collection

(Maximum word limit: 100 words) This project utilizes the LIDC-IDRI dataset obtained from The Cancer Imaging Archive (TCIA). The dataset consists of over 1,000 chest CT scans stored in DICOM format. The key statistics of the dataset are summarized as follows:

- **Total cases:** 1,018 patients
- **Total lesions:** 7,371 lesions marked as “nodule” by at least one radiologist
- **Nodules $\geq$ 3 mm:** 2,669 lesions identified as clinically significant nodules

lesions in three categories
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6. Expected Outcomes

(Maximum word limit: 100 words) Summarize the expected outputs (eq: demo, code files, presentation) and performance metrics (e.g., accuracy, clarity, speed). Explain how your results will demonstrate the effectiveness of your image processing or computer vision approach. Write the outcomes in point form.

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7. Percentage of Image Processing & Computer Vision Involvement

(Approximately 50–100 words) State clearly how your project satisfies the requirement of at least 65% image processing and computer vision components. For example, describe which parts involve preprocessing, segmentation, or model development etc. And if you are planning to use any tools may be you can mention.

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References

[1] R. Uppaluri, T. Mitsa, M. Sonka, *et al.*, "Quantification of pulmonary emphysema from lung computed tomography images," *American Journal of Respiratory and Critical Care Medicine*, vol. 156, no. 1, pp. 248–254, 1997.

[2] J. W. Song, J. G. Im, M. I. Ahn, *et al.*, "Ct findings for ground-glass attenuation in subacute hp," *Journal of Thoracic Imaging*, vol. 16, no. 3, pp. 234–242, 2001.

[3] K. Murphy, M. Prokop, and B. van Ginneken, "A hybrid method for segmentation of lung fields in chest radiographs," *Medical Image Analysis*, vol. 13, no. 2, pp. 327–334, 2009.

[4] Anonymous or representative author, "Markov random field-based segmentation for ground-glass opacities in lung ct," *Example Journal*, 2012. Representative citation for MRF-based GGO segmentation studies.

[5] Representative author, "Density mask techniques for ggo quantification in ct," *Example Journal*, 2005. Representative citation for density mask approaches.

[6] Representative author, "Semi-automated segmentation and radiomics of ggo in chest ct," *Example Journal*, 2021. Representative citation for semi-automated radiomics pipelines.

A minimum of ten (10) references must be included in the References section.